

Zero Day Threat Report

UniTrade v1 — Requirement Contradiction Analysis

Product: UniTrade Campus Marketplace **Version:** 1.0 Pre-Launch **Date:** April 27, 2026

Role: Lead Engineer

REFERENCE — FEATURE MATRIX

#	FEATURE NAME	DESCRIPTION
F1	Escrow Wallet Protection	Buyer payment is held inside UniTrade until the item is successfully handed over
F2	Instant Seller Payout	Seller receives money immediately when buyer taps Buy
F3	Anonymous Marketplace Mode	Users can browse and purchase without sharing personal profile details
F4	First Year Safety Restriction	First-year students cannot buy items above a fixed price threshold
F5	Final Sale at Handoff	Once the buyer accepts the item in person, the sale becomes final
F6	48 Hour Auto Refund Button	After pickup, buyers can trigger a no-questions-asked refund within 48 hours

#	FEATURE NAME	DESCRIPTION
F7	Verified Campus Email Sign Up	Every user must register with a valid university email
F8	Seller Rating and Flag History	Platform shows seller trust signals based on completed trades and prior reports

STEP 1 — TOXIC PAIRS IDENTIFIED

TOXIC PAIR 1

F1 — Escrow Wallet Protection

VS

F2 — Instant Seller Payout

Why they conflict: Escrow Wallet Protection requires that the buyer's payment is held by UniTrade until the physical handoff is confirmed. Instant Seller Payout requires that the seller receives funds immediately at the moment of purchase. These two rules impose mutually exclusive states on the same pool of money — it cannot simultaneously be held in escrow *and* released to the seller. If both features ship, the system has no deterministic answer to the question: *"Where does the money go the instant the buyer taps Buy?"*

TOXIC PAIR 2

F5 — Final Sale at Handoff

VS

F6 — 48 Hour Auto Refund Button

Why they conflict: Final Sale at Handoff declares the transaction **irreversible** the moment the buyer accepts the item in person. The 48 Hour Auto Refund Button grants the buyer a **48-hour window to reverse** that very same transaction after pickup. These two features define opposite post-handoff states: one says *"done forever"* and the other says *"still reversible for 48 hours."* They cannot both be the authoritative policy governing completed sales.

TOXIC PAIR 3

F3 — Anonymous Marketplace Mode

VS

F8 — Seller Rating and Flag History

Why they conflict: Anonymous Marketplace Mode promises that users can transact **without revealing their profile identity** to other students. Seller Rating and Flag History requires the platform to **publicly display a seller's trade history, ratings, and prior reports** — information that is inherently tied to a persistent, identifiable seller profile. If a seller is anonymous, there is no profile to attach ratings to. If ratings are displayed, the seller is no longer anonymous. The two features impose contradictory visibility rules on the same user entity.

EXPLOIT — TOXIC PAIR 1

Ghost Listing Cash-Out

A malicious seller lists a fake item (e.g., "MacBook Pro — ₹25,000"). A buyer taps **Buy**. Because Instant Seller Payout is active, the seller receives ₹25,000 **immediately**. The seller then never shows up for the handoff. Escrow was supposed to protect the buyer, but the money already left the system. The buyer has lost their money with no recourse, and UniTrade is liable for the gap.

RISK: DIRECT MONEY LOSS

EXPLOIT — TOXIC PAIR 2

Free-Rental Refund Loop

A buyer purchases a calculator before an exam. They accept the item at handoff, which the system marks as **Final Sale**. However, the 48 Hour Auto Refund Button is also live. After the exam, the buyer taps refund within 48 hours — no questions asked. The buyer used the calculator for free, the seller loses both the item and the money, and the platform's "Final Sale" guarantee is proven worthless. Repeat abusers can exploit this for every short-term need.

RISK: SYSTEM ABUSE & BROKEN TRUST

EXPLOIT — TOXIC PAIR 3

Flagged Seller Identity Reset

A seller accumulates multiple negative flags and a 1.5-star rating after selling defective items. Under Seller Rating and Flag History, buyers would see these warnings and avoid them. But with Anonymous Marketplace Mode active, the same seller simply lists items anonymously — their flag history and low rating are hidden. Buyers unknowingly purchase from a repeatedly reported seller. The trust system is completely bypassed, and the platform fails to protect its users.

RISK: TRUST SYSTEM BYPASS

STEP 3 — ENGINEER DECISIONS

DECISION — TOXIC PAIR 1

✕ **Remove: Instant Seller Payout**

Message to PM: We should remove **Instant Seller Payout** from V1 because the escrow protection flow — which is our primary buyer trust mechanism — cannot function if funds exit the platform the moment a purchase is made; buyer money must remain held until handoff confirmation to prevent fraud.

DECISION — TOXIC PAIR 2

✕ Remove: 48 Hour Auto Refund Button

Message to PM: We should remove the **48 Hour Auto Refund Button** from V1 because a no-questions-asked post-handoff refund directly invalidates our Final Sale at Handoff policy, creating a loophole that enables consequence-free item returns and systematic abuse by buyers.

DECISION — TOXIC PAIR 3

✕ Remove: Anonymous Marketplace Mode

Message to PM: We should remove **Anonymous Marketplace Mode** from V1 because allowing users to hide their identity renders the Seller Rating and Flag History system ineffective, enabling flagged or low-rated sellers to evade accountability and eroding the trust foundation of the marketplace.